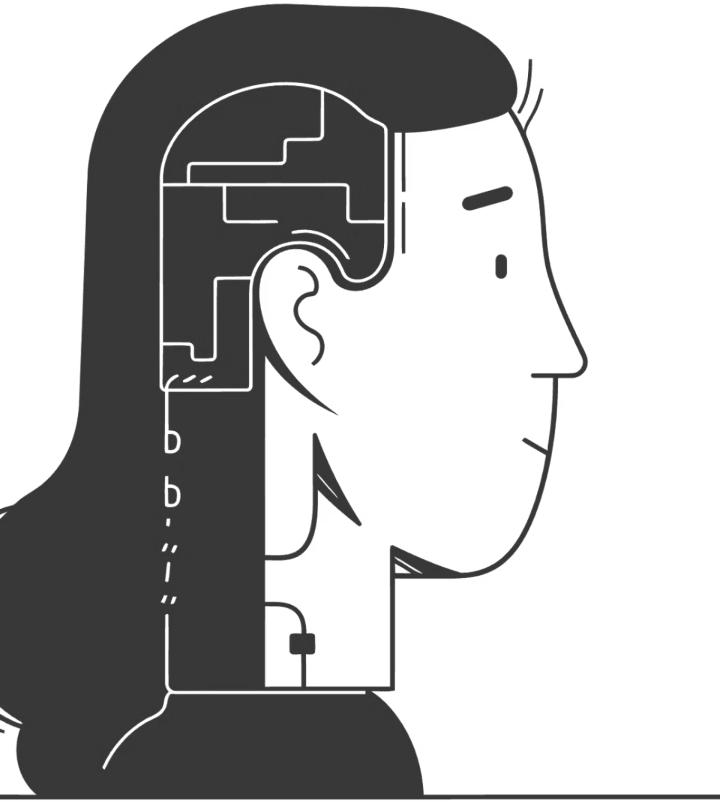


DATA-MD CURRICULUM

Session 2: Error and Bias in AI Systems

Data-Driven, Technology-Assisted Medical Decision Making for Medical Trainees

Emily Balczewski, PhD · Andrew Wong, MD, MS · Cornelius James, MD · Niharika Goparaju, MD · Thomas East, PhD · Joanne Baker, DO · Thalia Nguyen, MD · Serena Wang, MD · Lisa Zickuhr, MD

Pre-Test Assessment

Before we begin — let's benchmark your current understanding of AI in medicine.



Scan the QR code to complete the pre-test assessment. Your responses are anonymous and help us tailor the session.

Session 2 Learning Objectives

Identify Sources of Error & Bias

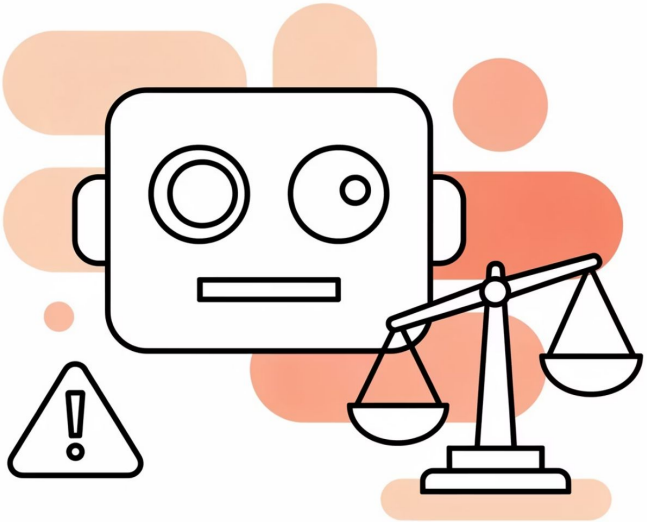
Recognize sources of error and bias in large language models (LLMs)

Prompt Engineering

Practice strategies to improve LLM outputs through better prompting

Small Group Activity

Using prompting techniques to generate a case report

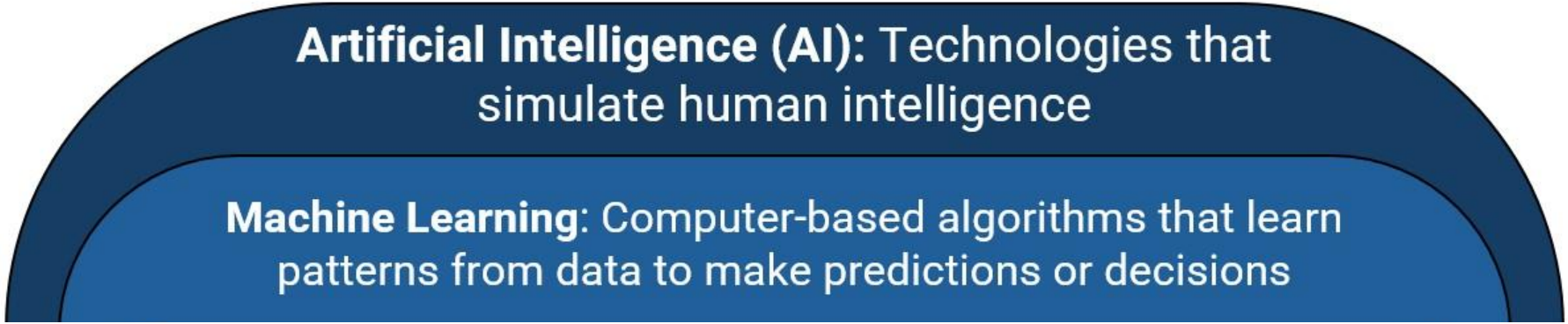


LOOKING BACK

Session 1 Review

Recap of foundational AI concepts: the AI landscape, machine learning basics, and how models learn from data.





Classical Machine Learning **vs.** **Deep Learning**

Linear Regression *Neural Network*

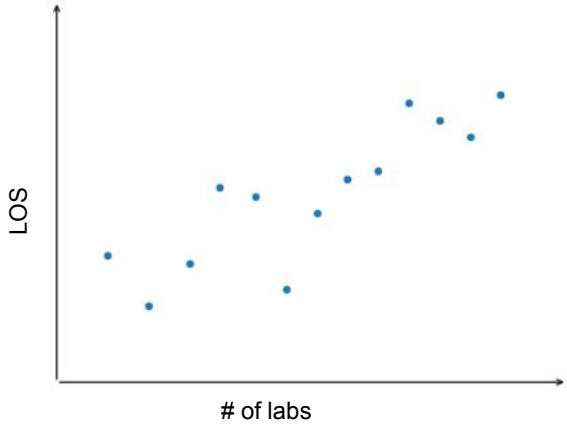
Artificial Intelligence (AI): Technologies that simulate human intelligence

Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Linear Regression

Predict ED LOS

1) Gather data
2) Train the model
3) Implement the model

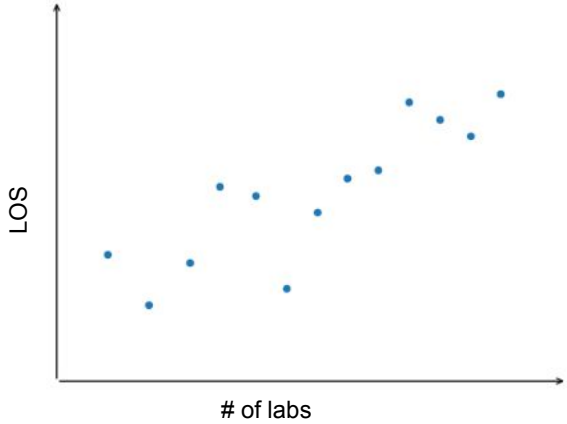


Artificial Intelligence (AI): Technologies that simulate human intelligence

Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Linear Regression
Predict ED LOS

1) Gather data
2) Train the model
3) Implement the model



Artificial Intelligence (AI): Technologies that simulate human intelligence

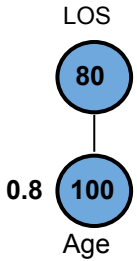
Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Linear Regression

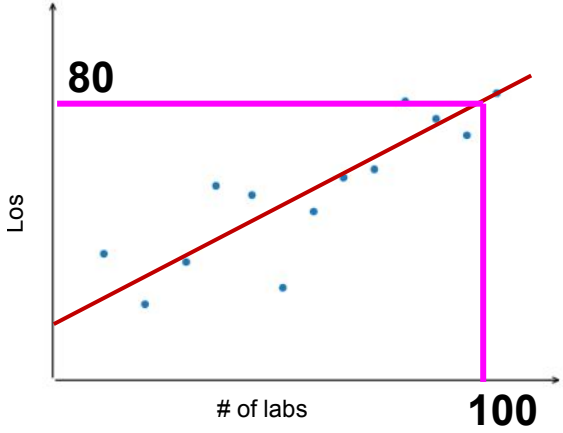
Predict ED LOS

y = m*x

LOS = 0.8*(# of Labs)



1) Gather data
2) Train the model
3) Implement the model



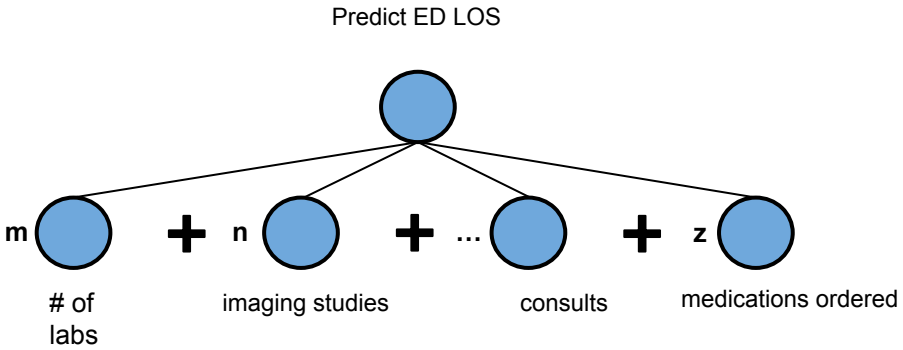
Artificial Intelligence (AI): Technologies that simulate human intelligence

Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Linear Regression

Predict ED LOS

+ labs + … + consults

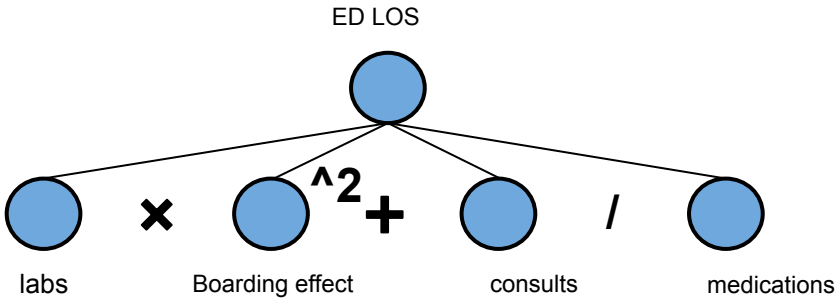


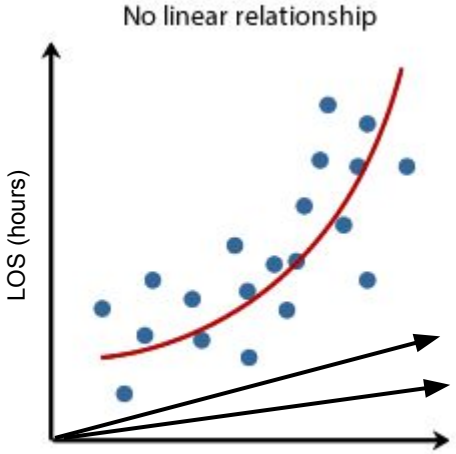
slope =
coefficient =
relationship between inputs and outputs
=
weights

Artificial Intelligence (AI): Technologies that simulate human intelligence

Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

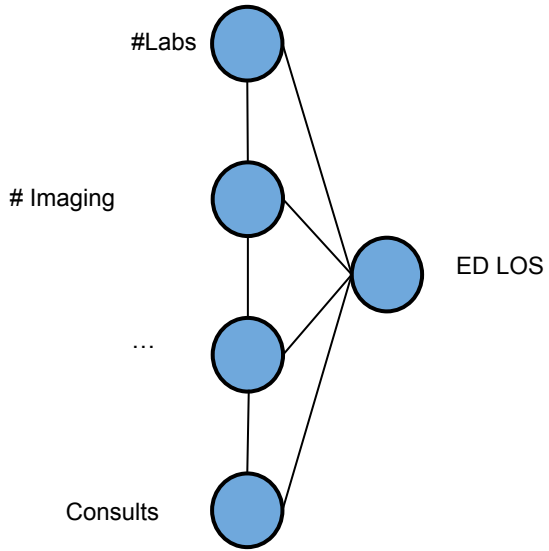
Nonlinear Regression
Predict ED LOS
* Boarding effect^2 + … / consult





Artificial Intelligence (AI): Technologies that simulate human intelligence

Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

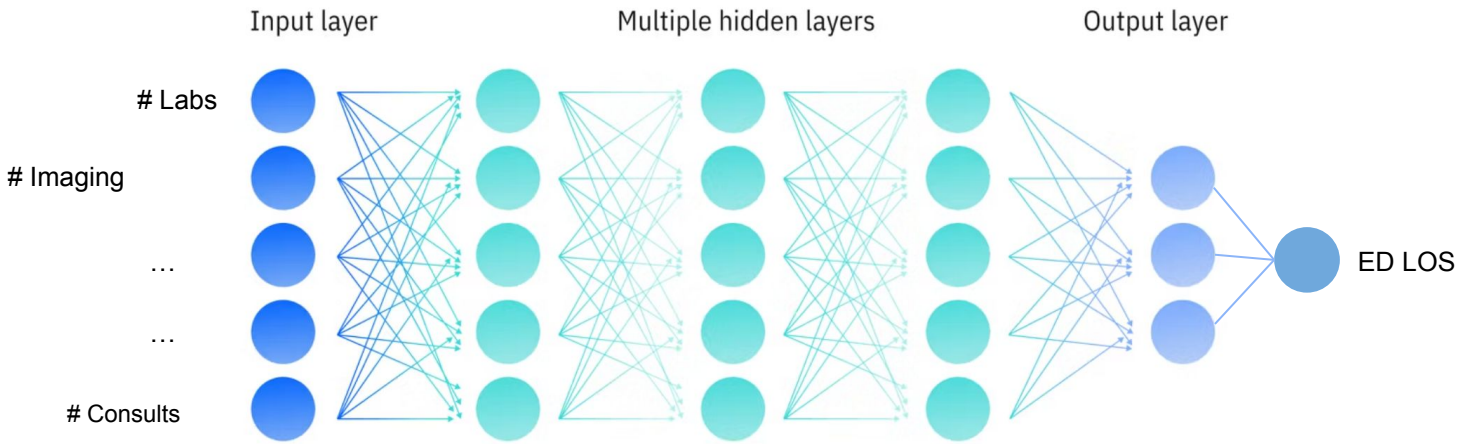


How do we get from Linear/Nonlinear Regression to Deep Learning?

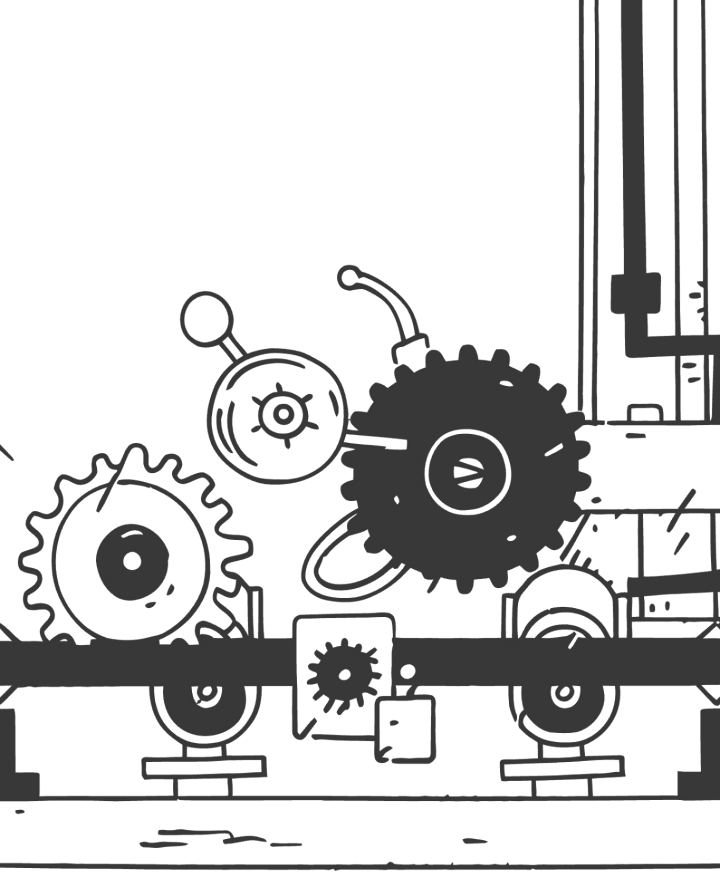
Artificial Intelligence (AI): Technologies that simulate human intelligence

Machine Learning: Computer-based algorithms that learn patterns from data to make predictions or decisions

Neural Network (Deep Learning)



ibm.com



KEY CONCEPT

Deep Learning = Layered Linear Regression

1

Linear Regression

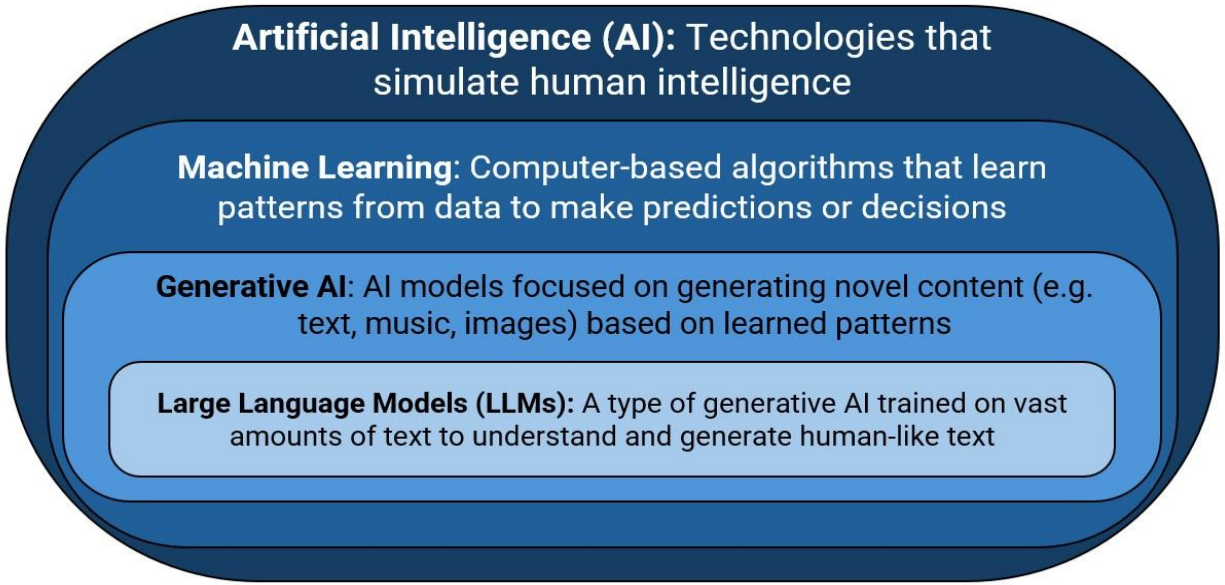
1 layer, interpretable

2

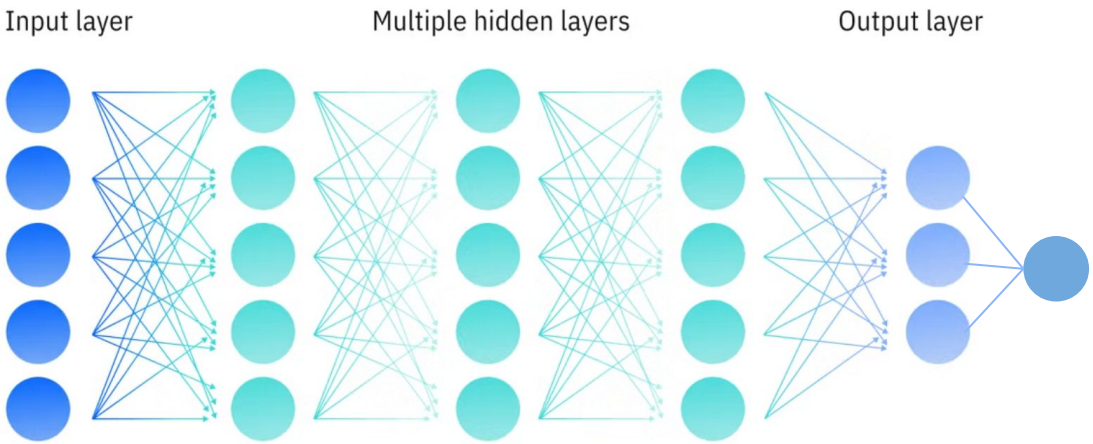
Deep Learning

Many layers, flexible, difficult to interpret

From Deep Learning to Large Language Models



Large Language Models (LLMs)



ChatGPT

OpenEvidence



LLAMA 3

Step 1: Tokenize

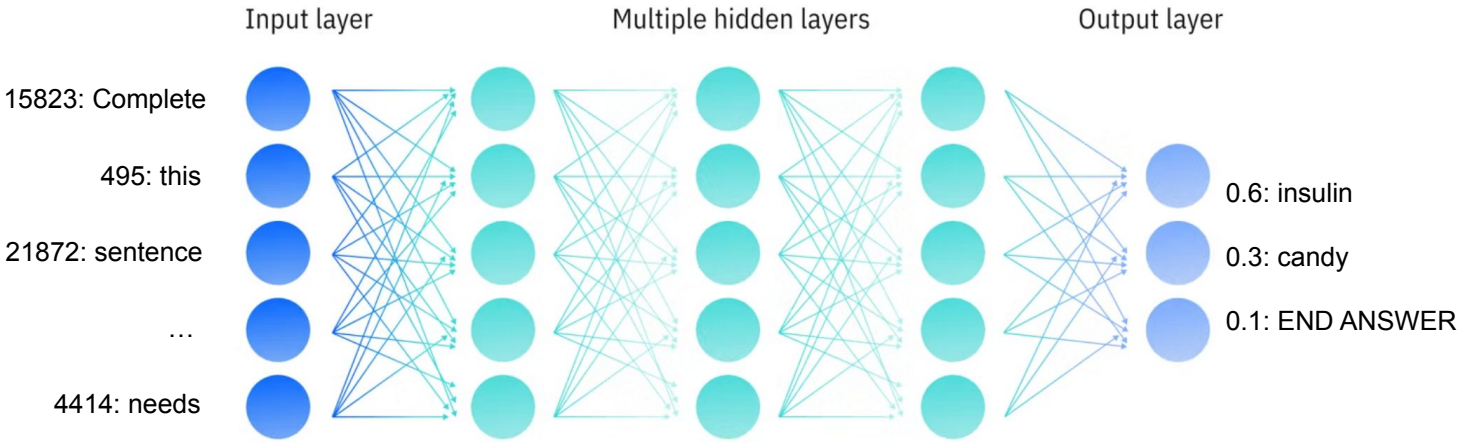
Tokens	Characters
11	63

> 💡 LLMs charge by the number of tokens in input **and** output — longer prompts = higher cost.

Complete this sentence: The patient with Type II Diabetes needs

[15823, 495, 21872, 25, 623, 10323, 483, 7003, 9839, 59910, 4414]

Step 2: Generate likelihoods for next token



Input: “Complete this sentence: The patient with Type II Diabetes needs”

Output:

How does an LLM get made?

1) Gather data: real and synthetic
2) Pretraining: learn the general statistical properties of language
    a) Self-supervised learning: learn how to do general tasks like predicting missing word in a sentence or predicting a missing pixel in an image
3) Supervised fine-tuning: learn how to perform a specific task, given examples of questions and answers
4) +/- Reinforcement learning: learn how to perform a specific task to the liking of humans, who rate different model outputs
5) +/- Extra features like Retrieval Augmented Generation
6) Deploy model: continuous cycles of oversight and monitoring

BLUEPRINT FOR AN

AI BILL OF RIGHTS

MAKING AUTOMATED SYSTEMS WORK FOR THE AMERICAN PEOPLE

OCTOBER 2022

THE WHITE HOUSE
WASHINGTON



Safe and Effective Systems



Algorithmic Discrimination Protections



Data Privacy



Notice and Explanation

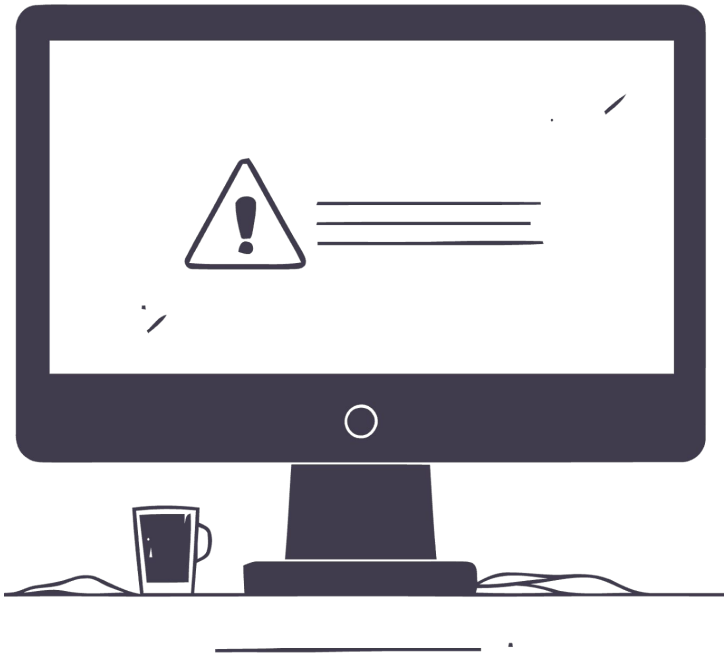


Human Alternatives, Consideration, and Fallback

SESSION 2 — NEW CONTENT

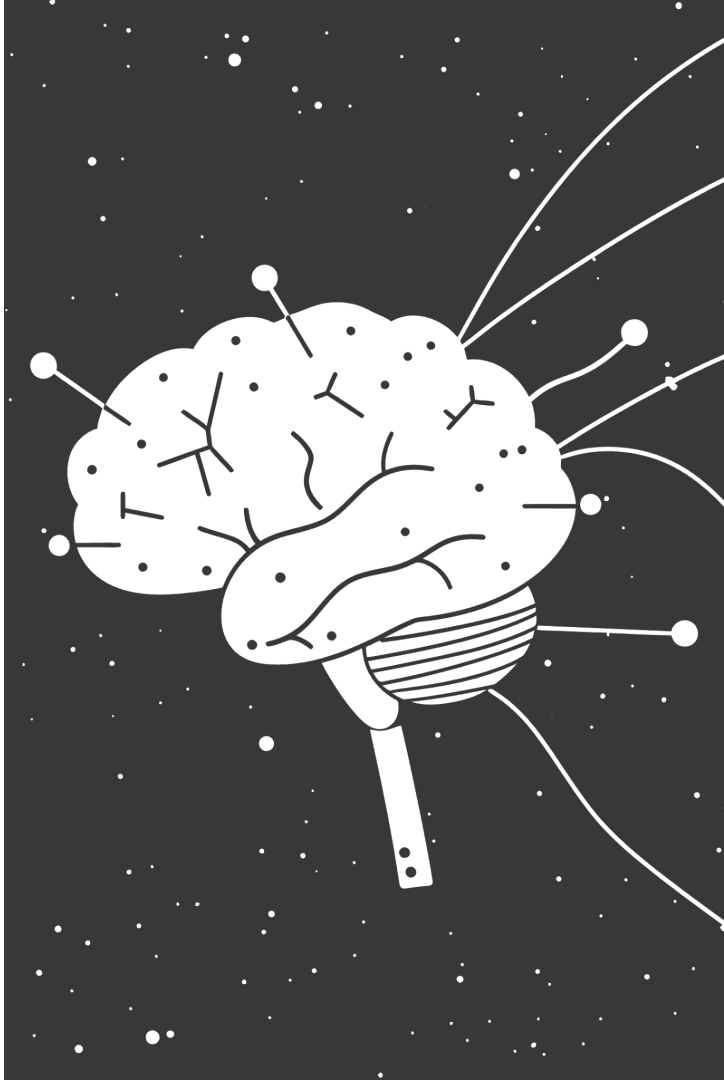
Error in AI Systems

Understanding where AI goes wrong is as important as understanding how it works — especially in high-stakes clinical settings.



Common Generative AI Sources of Error

4 categories of error



Error Type #1: Hallucination

1

Hallucination

Fabricated facts, sources, or data presented as fact

SUBTYPES

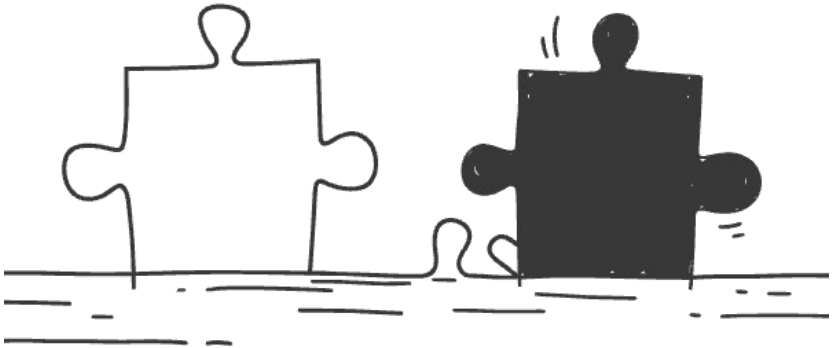
Inaccuracy

Structure of the info is correct, but a key detail is wrong.

Misattribution

Incorrect source cited, or a real source is misinterpreted

Error Type #2: Omission



Hallucination

Fabricated facts, sources, or data

Omission

Critical information excluded from the output

Error Type #3: Limited Reasoning



Hallucination

Fabricated facts, sources, or data



Omission

Critical information excluded from the output



Limited Reasoning / Logic

AI draws incorrect conclusions from information

Error Type #4: Text output error

1. Hallucination

Fabricated or unsubstantiated content

2. Omission

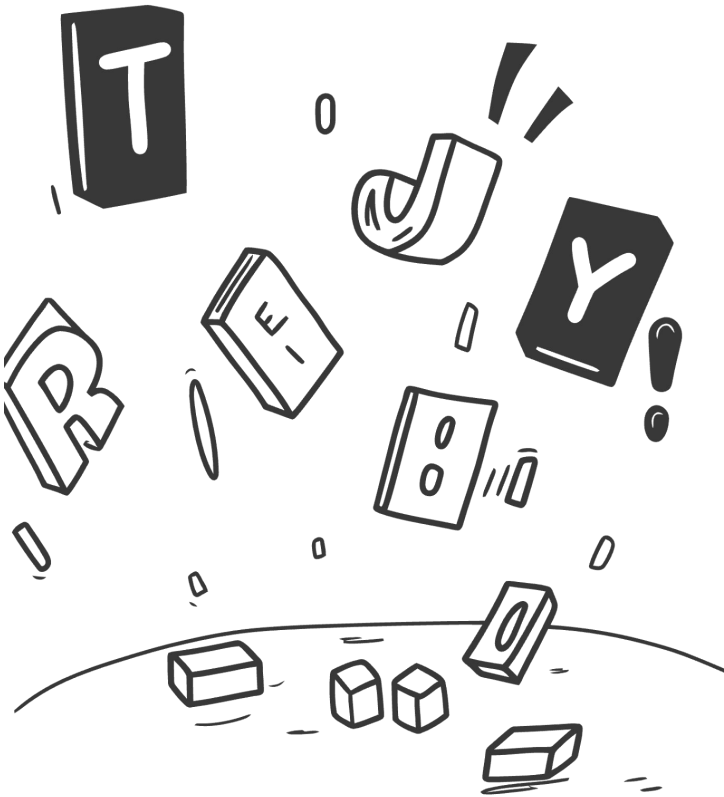
Critical information excluded

3. Limited Reasoning

Incorrect conclusions from correct data

4. Text Output Errors

Spelling, translation, grammar, fluency



Spot the Errors!

Clinical Prompt

A 16-month-old male presents after falling from a couch (~2 feet) onto wood floor. He cried immediately, had no LOC and has been acting normally since. Exam shows a small frontal scalp hematoma. Vitals are WNL.

What is the appropriate evaluation and management?



Spot the Errors!

Prompt

A 16-month-old previously healthy male presents after falling from a couch (~2 feet). He cried immediately, had no LOC and has been acting normally since. Exam shows a small frontal scalp hematoma. Vitals are WNL. **What is the appropriate evaluation and management?**

Read the AI output carefully — can you identify the errors?

LLM Output (Chat GPT)

"Thank you so much for providing such a detailed and thoughtful clinical vignette!"

Recommended management:

- Obtain a head CT to rule out intracranial injury
  - Discharge home if CT is normal
- Risk of clinically significant brain injury is *moderate* due to age
- Frontal scalp hematomas are concerning for intracranial bleeding.
  - Smith J et al. *J Pediatr Emerg Med*, 2022

Great job recognizing a potentially serious head injury!

Errors Identified

AI Output — Annotated

Hallucination

1. **Misattribution:** Smith J et al. is a fake source.
2. **Inaccuracy:** Frontal scalp hematomas are concerning

Omission

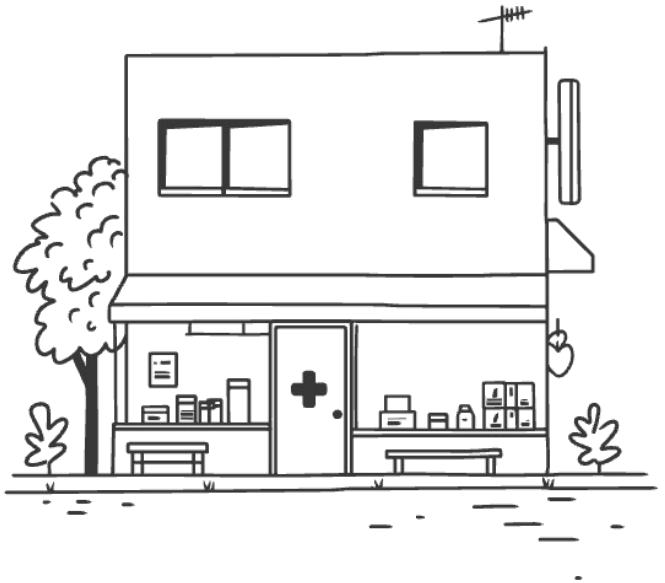
Missing: Use of Clinical decison rule —> (PECARN) and if anything: observation is a primary strategy.

Limited reasoning

- No risk stratification
- Jumping straight to CT (doesn't weigh radiation risk or pretest probability)

Sycophancy Bias

Excessive flattery masks errors and erodes critical evaluation



Discussion Question

Which AI errors are hardest to catch?

Reflect on your own experience using AI tools in clinical or educational settings.



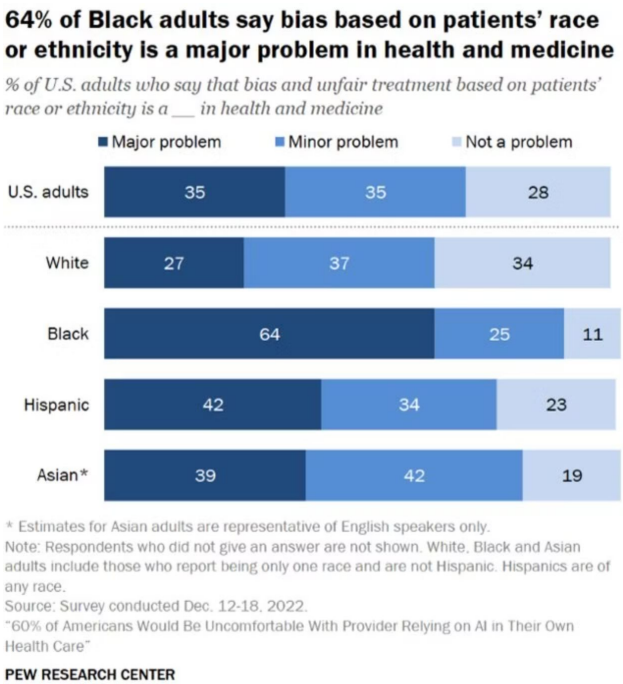


NEW TOPIC

Bias in Generative AI

AI systems reflect the data they were trained on — and the biases embedded within it.

⚠ Bias ↑ → Generalizability ↓ → Equity ↓



Why Does Bias in AI Matter?

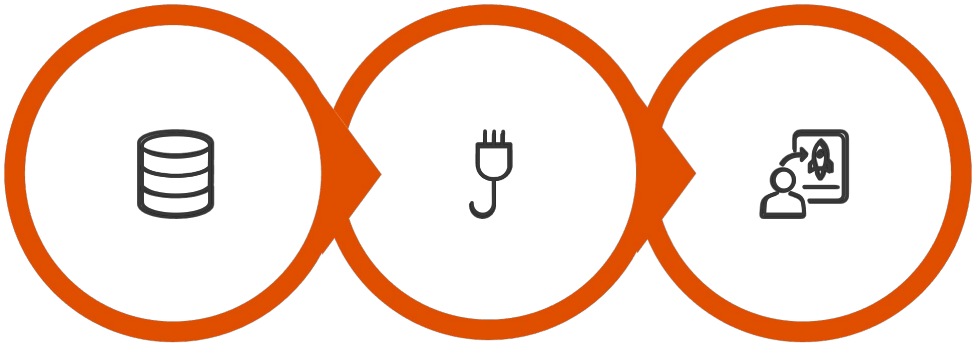
64% of Black adults say race/ethnicity-based bias is a **major problem** in healthcare.

↑ Bias
AI reflects flawed training data

↓ Generalizability
Performs poorly across populations

↓ Equity
Widens existing health disparities

Bias Can Arise at Every Stage



Gather Data

Train Model

Implement Model

No stage is immune — bias introduced early compounds at every subsequent step in the pipeline.

STAGE 1

Bias at Data Collection

1

Representation Bias

Training data under-represents certain demographic subgroups

Ex. LLMs performs worse on translation tasks in less common languages

2

Temporal Bias

Time-based factors skew model outputs

Ex. A model trained on 2010–2015 data underestimates COVID-19 hospitalization risk in 2020

STAGE 2

Bias During Model Training

Algorithmic Bias

Model learns the wrong patterns because of the variables given (by biased data or design)

Evaluation Bias

Errors in how a model is tested or validated, leading to misleading performance conclusions

Ex. A self-driving AI drives perfectly in clear weather, but stops in snow due to over-perceived danger

Clinical Risk

Biased models may recommend inferior care for underrepresented patients

STAGE 3

Bias at Implementation

Deployment Bias

Theoretical model use differs from real-world implementation

Ex. A remote monitoring algorithm using wearables performs poorly in rural populations with limited internet access

Key Takeaway

Even a well-trained model can fail if implemented in a context it wasn't designed for.

⚠ Always evaluate whether the deployment context matches the training context.

Sycophancy: A Bias That Permeates LLMs

AI models are **trained to please users** via reinforcement learning — they prioritize responses users will approve of, even over accuracy.



→ Over-agreeing

Tells you what you want to hear

→ Flattering responses

Excessive praise lowers your critical guard

→ Avoiding difficult truths

Biased or inaccurate answers to avoid conflict

Mitigating AI Pitfalls

Verify All Outputs

Cross-check AI-generated content against trusted clinical sources

Use RAG / Trusted Sources

Integrate verified references directly into your prompts

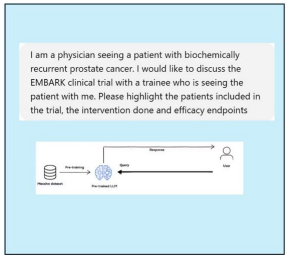
Follow Legal & Ethical Principles

Assume your prompts are recorded and surveilled

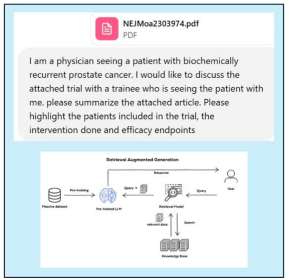
Optimize Prompt Engineering

Craft precise, context-rich prompts to reduce error

You can improve output by providing the data...



vs



Providing source documents (RAG) dramatically improves output accuracy and reduces hallucination.

Prompt Engineering for Healthcare

Mastering AI communication for better clinical outcomes



ChatGPT

Ask anything

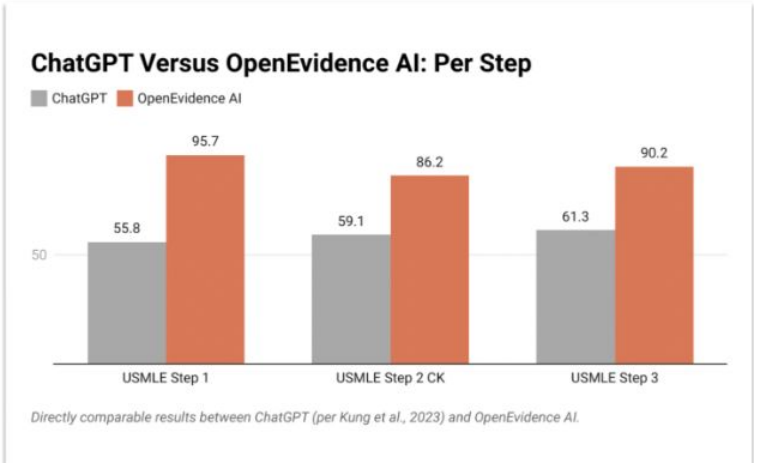
Attach Search Study Voice

OpenEvidence

Ask a medical question...

Write Home Care Instructions NEW | Ask for a Table | Ask for Evidence

Explore More Capabilities



Source: openevidence.com

Generating a good prompt

1. Be Clear and Specific
2. Frame Your Context
3. Experiment with Iterative Questions
4. Limit Potential Bias
5. Analyze and Adapt

learnprompting.org

But… are there any other techniques that are useful?

TECHNIQUE 1

Meta-Prompting

Ask the LLM to design the prompt

Have the LLM help you craft, refine, or interpret a prompt — rather than just completing the task.

Clinical Example: "I'm a surgeon writing discharge instructions for a post-op appendectomy patient. Please improve my prompt to create patient-friendly, clinically appropriate instructions."



Improved Prompt:

You are a clinician writing discharge instructions for a patient recovering from an appendectomy. Create a clear, compassionate, and clinically accurate set of instructions written at a 6th-grade reading level.

Include the following sections:

1. "**Overview:** Briefly explain what an appendectomy is and what the patient can expect after surgery."
2. "**Wound Care:** Step-by-step guidance on caring for the incision (e.g., dressing changes, keeping area dry, when to shower)."
3. "**Activity:** Guidance on walking, lifting restrictions, return to work/school, and resuming exercise."
4. "**Diet:** Advice on advancing diet, managing nausea, and hydration."
5. "**Pain Control:** Instructions for using pain medications safely (include both OTC and prescribed meds)."
6. "**Bowel Movements:** Address constipation prevention after anesthesia and pain meds."
7. "**Warning Signs:** List of symptoms that should prompt contacting the surgeon or visiting the ER (e.g., fever, redness, drainage, worsening pain)."
8. "**Follow-Up:** Include when and how to follow up with the surgical team."
9. "**When to Seek Immediate Help:** Clear, bolded section with emergency symptoms."

Keep language supportive and reassuring. Avoid medical jargon. If possible, include a short summary at the end reminding the patient that recovery takes time and to contact their care team with any concerns.

TECHNIQUE 2

Self-Criticism Prompting

Have the model evaluate its own output

Powerful for QA, research writing, and clinical teaching.

Prompt 1

"*Please give a broad outline of objectives for teaching a simulation to residents on: An adult patient with respiratory distress due to carbon monoxide poisoning.*"

Follow-Up Prompt

Refine this list — remove low-yield items, add missing ones relevant to teaching goals: *"recognition, stabilization, decision-making and teamwork."*

🚫 Objectives Removed (Low-Yield for This Case)

❌ Detailed pathophysiology (e.g., hemoglobin binding curve shifts) — useful for lecture, not simulation.

❌ Extensive discussion of delayed neuropsychiatric sequelae — can be post-sim teaching point, not in-scenario.

❌ Public health reporting — not primary to learner decision-making focus.

❌ Documentation objectives — lower priority for a crisis-oriented simulation.

❌ General differential for respiratory distress (e.g., COPD, asthma) — too broad for targeted CO sim.

🔍 Key Additions Made

✅ Emphasis on **recognizing false reassurance from SpO_2**

✅ Integration of **team leadership and task prioritization**

✅ Inclusion of **scene safety and environmental hazard identification**

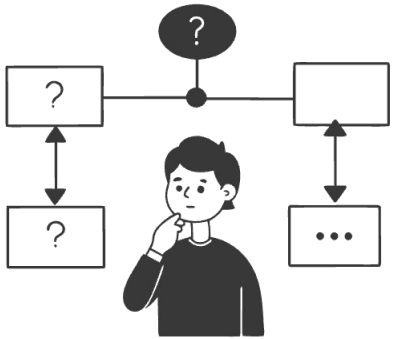
TECHNIQUE 3

Chain of Thought Prompting

Concept: Encourage the model to show its work – particularly useful for diagnostic reasoning, guideline application, or risk stratification.

Clinical Example:

Walk through step-by-step reasoning for calculating a clinical HEART SCORE in a 67-year-old male with a past medical history of HTN and prior MI in 2018?



HEART score: how to score each element

- **H — History (0–2)**
  - 0: Slightly suspicious for ACS
  - 1: Moderately suspicious
  - 2: Highly suspicious
  - *Use the story: character of pain, radiation, exertional nature, diaphoresis, similar to prior MI, etc.*
- **E — ECG (0–2)**
  - 0: Normal
  - 1: Nonspecific repolarization abnormalities (e.g., ST/T changes not meeting ischemia)
  - 2: Significant ST depression (new) ± pathologic T-wave inversion
  - *Do not give points for isolated LVH or paced rhythm unless ischemic changes present.*
- **A — Age (0–2)**
  - 0: <45 years
  - 1: 45–64 years
  - 2: **≥65 years** ← *your patient gets 2 here*
- **R — Risk factors (0–2)**
  - Traditional risks: HTN, HLD, DM, obesity, current smoking, positive family history (premature CAD).
  - **Known atherosclerotic disease** (prior MI, PCI/CABG, stroke/TIA, PAD) counts as "high."
  - 0: No risk factors
  - 1: 1–2 risk factors
  - **2: ≥3 risk factors OR known atherosclerotic disease** ← *prior MI ⇒ 2 here; HTN also present*
- **T — Troponin (0–2)**
  - 0: ≤ upper reference limit (URL) for your assay
  - 1: >URL to ≤3× URL
  - 2: >3× URL
  - *Use your lab's ULN/URL; with hs-cTn, apply the same concept using the assay's cutoffs.*

Apply to this patient

- **Age:** 67 → **2 points**
- **Risk factors:** Prior MI (known CAD) ± HTN → **2 points** (automatically high by HEART rules)
- **History:** depends on the presentation (0–2)
- **ECG:** depends on today's tracing (0–2)
- **Troponin:** depends on the value relative to your assay's URL (0–2)

Fixed subtotal = 4 (Age 2 + Risk 2).
Final HEART = 4 + (History 0–2) + (ECG 0–2) + (Troponin 0–2) → **range 4–10** for this patient.

Common scenarios for this case

- **Best-case (still not "low-risk" due to age/CAD):**
  History slightly suspicious (0) + ECG normal (0) + Troponin normal (0) → **HEART = 4** (moderate risk).
- **Typical concerning case:**
  Moderately suspicious history (1) + nonspecific ST/T changes (1) + Troponin just above URL (1) → **HEART = 7** (high risk).
- **Clearly high-risk case:**
  Highly suspicious history (2) + ST depression (2) + Troponin >3× URL (2) → **HEART = 10** (very high).

Interpreting the score (original HEART data)

- **0–3 (low):** ≈1–2% 30-day MACE → often suitable for early discharge if serial hs-troponins negative.
- **4–6 (moderate):** ≈12–16% → observe, serial troponins, further testing per local pathway.
- **7–10 (high):** ≈50–65% → urgent/early invasive evaluation.

Because your patient starts at **4** before history/ECG/troponin, they will almost always fall into **moderate or high risk** once the other elements are added.

TECHNIQUE 4

Brevity*

Concept: Easier to validate an output if it is *short*.

Clinical Example:

Walk through step-by-step reasoning for calculating a clinical HEART SCORE in a 67-year-old male with a past medical history of HTN and prior MI in 2018? Be brief.

Minimum Known Score So Far

- **Age: 2**
- **Risk factors: 2**
- **Subtotal = 4 points**

Final HEART score will range **4–10**, depending on:

- History (0–2)
- ECG (0–2)
- Troponin (0–2)

Interpretation

- **0–3**: Low risk
- **4–6**: Moderate risk
- **7–10**: High risk

👉 With known data alone, this patient is **already at least moderate risk**.

HEART score: how to score each element

- **H — History (0–2)**
  - 0: Slightly suspicious for ACS
  - 1: Moderately suspicious
  - 2: Highly suspicious

  Use the story: character of pain, radiation, exertional nature, diaphoresis, similar to prior MI, etc.
- **E — ECG (0–2)**
  - 0: Normal
  - 1: Nonspecific repolarization abnormalities (e.g., ST/T changes not meeting ischemia)
  - 2: Significant ST depression (new) ± pathologic T-wave inversion

  Do not give points for isolated LVH or paced rhythm unless ischemic changes present.
- **A — Age (0–2)**
  - 0: <45 years
  - 1: 45–64 years
  - 2: **≥65 years** ← *your patient gets 2 here*
- **R — Risk factors (0–2)**
  - Traditional risks: HTN, HLD, DM, obesity, current smoking, positive family history (premature CAD).
  - **Known atherosclerotic disease** (prior MI, PCI/CABG, stroke/TIA, PAD) counts as "high."
  - 0: No risk factors
  - 1: 1–2 risk factors
  - **2: ≥3 risk factors OR known atherosclerotic disease** ← *prior MI ⇒ 2 here; HTN also present*
- **T — Troponin (0–2)**
  - 0: ≤ upper reference limit (URL) for your assay
  - 1: >URL to ≤3× URL
  - 2: >3× URL

  Use your lab's ULN/URL; with hs-cTn, apply the same concept using the assay's cutoffs.

Small Group Activity (15 minutes)

Practicing Clinical Decision Making with Large Language Models



Scenario: 4-yr-old male · Six days of fever · Two outpatient visits before admission · Complete KD criteria met day 6 · Mild bilateral coronary dilation on echo.

In groups of 2-3:

1. Please write a full *Pediatrics* Case Report: ⩽250-word unstructured abstract + ⩽1,600-word manuscript
2. Pick a model or several models
3. Run different prompts and compare your techniques and outputs

Generating a good prompt

1. **Be Clear and Specific**
2. **Frame Your Context**
3. **Experiment with Iterative Questions**
4. **Limit Potential Bias**
5. **Analyze and Adapt**

Advanced Prompt Strategies:

- **Meta-Prompting**
- **Self-Criticism Prompting**
- **Chain-of-Thought Prompting**
- **Brevity***



Report Out

How did it go? Share your experience with the group.

1. What tool(s) did you use?

2. What prompt did you write? How would you rate the output? What surprised you?

3. Would you trust these tool in clinical practice? Has it ever changed your management?

CLINICAL WRITING WORKFLOW · APRIL 2026

From case packet to journal-ready manuscript

A three-skill pipeline for a CARE-compliant Kawasaki disease case report

clinical-research-planning

clinical-paper

docx

The three-skill pipeline

Each skill consumes one set of files and produces another. Files persist between skills, so the work is reviewable at every stage.

01

PLAN

clinical-research-planning

Inventory inputs, scout literature, decide angle and audience.

INPUTS

- Case packet PDF
- AHA guideline (Jone 2024)
- Texas Children's pathway

OUTPUTS

- brief.md (drafting plan)
- research/library.bib (11 refs)
- research/notes/*.md
- submission/target-journal.md

02

DRAFT + VALIDATE

clinical-paper

Draft section-by-section in citekey syntax, plan figures, run the validation suite.

INPUTS

- brief.md (approved)
- library.bib
- Case packet PDF

OUTPUTS

- manuscript/paper.md
- manuscript/paper.vancouver.md
- figures/fig1-timeline.png
- figures/table{1,2}-*.md
- Plagiarism + PHI reports

03

EXPORT

docx

Render the Vancouver-numbered draft as a journal-ready Word document.

INPUTS

- paper.vancouver.md
- Figure 1 PNG
- Table 1 + Table 2 sources

OUTPUTS

- manuscript/paper.docx
- manuscript/paper.pdf

01 · clinical-research-planning

Decide what the paper will argue, before any drafting begins

WHAT THE SKILL DID

Six steps, one approval gate

1. **Inputs —** Read case packet + AHA + Texas Children's; PHI scan.
2. **Elicit —** Asked angle, audience, journal, take-away, consent via inline form.
3. **Search —** Targeted PubMed search anchored on diagnostic-delay literature.
4. **Synthesize —** Grouped 11 references into four themes; named the gap.
5. **Plan —** Wrote the abstract + manuscript drafting plans as task lists.
6. **Approve —** Required author sign-off before downstream skills run.

THE OUTPUT — brief.md

A drafting contract, not a research summary

- Approval status (gates the next skills)
- What the manuscript will argue
- Audience and target journal
- De-identified case summary
- Ranked teaching points
- Literature review by theme + gap statement
- Proposed structure (abstract + paper)
- Drafting plan — task list per skill
- Self-review & validation rules
- Open questions for the author (7 deferred)

02 · clinical-paper

Draft section by section, validate before the author reads it

1,589
manuscript words / 1,600 cap

10
Vancouver-numbered references

1 + 2
figure + tables (CARE timeline + labs + criteria)

VALIDATION SUITE — every check ran before the author saw the draft

Check	Status	Note
Word count ≤ 1,600	PASS	1,589 / 1,600
CARE conformance (13 items)	PASS	Abstract item deferred to clinical-abstract; patient perspective records guardian declined
Figure / table integrity	PASS	Figure 1 + Tables 1–2 referenced; legends drafted; de-ID confirmed
Citation integrity	PASS	10/10 cited citekeys resolve; 0 NEEDS-REF placeholders
Vancouver export	WARN	Texas Children's institutional-author edge case in references list — single-line hand-fix flagged
PHI scrub	PASS	No initials, MRN-pattern numbers, dates, or institution names in body or figure sources
Plagiarism self-check	PASS	0 HIGH, 0 MEDIUM, 2 INFO false positives on patient-specific clinical numbers
Internal consistency (paper)	PASS	Numbers, z-scores, illness-day refs all consistent within the paper

THE FINAL OUTPUT

manuscript/paper.pdf

13-page docx · **1,589** words · **10** references · **1** figure · **2** tables · CARE-compliant · plagiarism PASS · PHI PASS



Key Concept

Do not put PHI in non-HIPAA-compliant LLMs.

What is PHI?

Any information that could identify a patient — name, DOB, MRN, diagnosis linked to identity, images, and more.

Safe Practice

- Use **de-identified, hypothetical** scenarios for learning
- Only use PHI in **HIPAA-compliant, institution-approved** platforms
- When in doubt — **ask your compliance office**

AI-AUGMENTED MEDICINE

The Skills You'll Need More Than Ever



Core Competencies for Medical Students, 2024

> The future physician isn't just a knowledge retainer — but a master of reasoning, empathy, and judgment.

Evidence-Based Medicine

Applying rigorous evaluation to all sources — including AI outputs — to ensure recommendations are grounded in the best available evidence.

Critical Evaluation of AI Tools

Assessing the validity, potential bias, and clinical applicability of AI-generated results before acting on them — never accepting outputs at face value.

Clinical Reasoning

Developing and refining differential diagnoses independently, using AI as a supportive resource rather than a substitute for sound medical thinking.

Course Outline

Session 1: Introduction to AI

- What is AI? Why do we care?

Session 2: Error and Bias in AI Systems

- AI for Clinical Decisionmaking

Session 3: Evaluating and Appraising AI Systems

- AI Benchmarks; Critical Appraisal

Session 4: Clinical Applications of AI

- AI for Documentation; Critical Appraisal
- Future Directions of AI

